

Aldrick Victor Tadeo Arellano

+1 (702) 624-4515 | aldricktadeo@gmail.com | github.com/aldrick-t | linkedin.com/in/aldrick-t

Robotics and embedded software engineer focused on ROS2-based autonomous systems, perception, SLAM, and sensor fusion. Experienced in developing real-time telemetry, computer vision pipelines, embedded firmware, and robotic control software across academic, competition, and industry environments.

Education

Bachelors of Science, Robotics and Digital Systems Engineering <i>Monterrey Institute of Technology and Higher Education (ITESM)</i>	2022 — 2026 GPA: 4.0/4.0
- Academic Excellence Scholarship recipient. - Specialization in Cyberphysical Systems	

Experience

Software Engineer Intern, GE Aerospace	Nov 2025 — Present
- Designed a usage-telemetry framework for multiple internal CAD/CAE (Siemens NX) and PLM/data tools; defined event model + PostgreSQL schema for hundreds of daily users. - Integrated telemetry into engineering toolchains via a C#/.NET service wrapper, sending events through REST + internal service calls. - Developed NX Open plugins to employ NX workflows with internal tools and capture usage/error data with minimal impact to engineers. - Built Grafana dashboards to track adoption, workflow bottlenecks, and reliability trends across CAD/PLM tooling.	
Telemetry Development Division Lead & Founder, Electrum Performance Racing Team	Jun 2024 — Oct 2025
- Founded and led the Telemetry Division, building a robust real-time data acquisition system designed for high-stress environments. - Developed failsafe telemetry solutions using serial communication protocols, reducing transmission failures by 15%. - Improved cross-team collaboration by implementing Fusion 360's cloud-based version control, cutting design iteration times by 10%. - Directed a cost optimization initiative, reducing manufacturing and component expenses by 35% while maintaining system reliability. - Engineered firmware components to streamline workflows and enhance data accuracy across multiple vehicle subsystems.	

Projects

Lead Developer, LiDAR Based SLAM for Autonomous Navigation	Mar 2026 — Present
- Developed a ROS2 LiDAR SLAM pipeline for autonomous navigation using scan matching and pose graph optimization, targeting real-time execution on embedded compute hardware. - Implemented a custom SLAM algorithm combining scan matching and graph optimization, improving localization accuracy by 20% compared to traditional methods. - Integrated using ROS2, enabling seamless communication with other robotic components and facilitating multi-sensor fusion.	
Robotic Design & Software Lead, IEEE Robosoft Competition 2026	Aug 2025 — Apr 2026
- Designed a soft robotic gripper for berry harvesting, including the tool, control software, and circuitry; optimized through iterative design to reduce material costs and enhance flexibility. - Developed IoT-based crop monitoring hardware and software for greenhouse environments, improving real-time tracking of crop health. - Built an AI vision system for in-field strawberry and raspberry classification, optimized for deployment on both remote servers and edge devices. - Created an object localization system using an in-hand RGB-D camera to guide robotic tool positioning with precision	
AI Navigation Systems Developer, Manchester Robotics & Tec de Monterrey	Feb 2025 — Aug 2025
- Developed autonomous navigation and control systems on the MCR2 Puzzlebot platform, enabling robust mobile robotics operation. - Implemented traffic sign and traffic light recognition systems, improving reliability, accuracy, and environmental adaptability by 33%. - Enhanced navigation and positioning precision by fusing visual and internal sensors for improved real-world performance. - Built vision redundancy features, reducing false positives and lowering detection/decision error rates by 15%. - Redesigned the ROS2 node-topic architecture, optimizing data throughput and minimizing system failure points.	

Skills

Programming: Python, C/C++, C#, MATLAB, JavaScript, VHDL
Robotics & Autonomy: ROS2, SLAM, RGB-D cameras, LiDAR, Robot Control
Embedded & Hardware: Circuit Design, KiCAD, FPGAs, ESP32, STM32, Arduino, Raspberry Pi
AI & Perception: OpenCV, Machine Learning, Deep Learning, Computer Vision, Object Detection
Tools: Linux, Docker, Grafana, PostgreSQL, Siemens NX, SolidWorks, Fusion 360
DevOps: Git/GitHub, CI/CD, Azure DevOps, Agile/Scrum, Unit Testing
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Certificates

Fundamentals of Deep Learning, NVIDIA	Credential ID: 1tO0Ys3ITkGJkXM3sgBKrQ
Generative AI with Diffusion Models, NVIDIA	Credential ID: TauXuWfURMOBYNnutOVkopw
OpenCV Bootcamp, OpenCV University	Credential ID: 65a25e083f50497dba5f5538026087

